
A Cross-Sectional Factor Library

for the Vietnamese Equity Market

Thirty alpha factors across seven families, with microstructure-aware construction and Vietnam-specific empirical calibration

Scope. This note specifies a working catalogue of thirty cross-sectional equity factors intended for systematic, market-neutral, or long-only tilt strategies on the HOSE, HNX, and UPCOM exchanges. Each factor is given a precise definition, an economic rationale, an expected cross-sectional sign, the data required to compute it, and—critically—a calibration note that flags how the factor has historically behaved in Vietnam, which differs in several important cases from the developed-market consensus.

Intended reader. A quantitative researcher or portfolio manager building or stress-testing a Vietnam equity model. Familiarity with the Fama–French factor framework, information coefficients, and standard portfolio-sort methodology is assumed.

Status of the market. As of mid-2026, Vietnam sits at an inflection point: FTSE Russell confirmed its reclassification from Frontier to Secondary Emerging Market in April 2026, with index inclusion phased in from 21 September 2026. This changes the marginal-investor mix and therefore the expected return to several of the factors below. The note is written with that transition in mind.

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1 Executive summary

This document catalogues thirty cross-sectional factors for the Vietnamese equity market, organised into seven families: **value** (6), **quality and profitability** (6), **investment and growth** (2), **momentum and reversal** (4), **risk and volatility** (4), **liquidity and size** (4), and a final family of **Vietnam-specific flow and microstructure** factors (4).

The reason a Vietnam-specific catalogue is warranted—rather than a direct import of a US or pan-European factor set—is that several anchor anomalies behave differently in this market. Three points summarise the empirical landscape and recur throughout the note:

1. **Short-term reversal dominates; cross-sectional momentum is fragile.** The one-month return reversal is one of the most robust regularities in Vietnamese data, while twelve-minus-one momentum is regime-dependent—it tends to appear only after up-markets and is consistent with behavioural overreaction rather than a stable risk premium. A naive long-momentum sleeve imported from developed markets is therefore not a safe default here.
2. **Value is real but contested, and “growth” has paid in places.** The Fama–French five-factor model out-explains the three-factor model in Vietnam and the value factor (HML) remains non-redundant; operating profitability is the best-performing quality proxy. Yet some studies using point-in-time data find that liquid, growth-oriented large caps have outperformed over particular windows—a consequence of foreign flow concentrating in a narrow set of investable names.
3. **Liquidity and size effects are weak or inverted.** The textbook illiquidity premium is muted and sometimes reversed: because foreign capital crowds into the most liquid stocks, illiquid small caps have not reliably outperformed. Size as a standalone factor is weak. This makes liquidity primarily a *risk and capacity* control in Vietnam rather than a dependable alpha source.

A fourth theme is unique to Vietnam and has no clean developed-market analogue: **foreign ownership room** and **foreign net flow**. Statutory foreign ownership limits (notably 30% for banks) mean that for “room-full” names, foreigners cannot be the marginal price-setter, while net foreign buying is itself a tradable signal. The final factor family formalises these.

The body of the note proceeds as follows. Section 2 sets out the market and microstructure constraints that govern what is actually implementable. Section 3 fixes the methodology—universe, standardisation, neutralisation, sign convention, and evaluation. Section 4 is the factor catalogue. Section 5 synthesises the Vietnam evidence into a single comparative table. Section 6 covers composite construction, capacity, and transaction costs. Section 7 lists implementation pitfalls specific to this market.

2 The Vietnamese equity market: structure and microstructure

2.1 Exchanges, size, and indices

Vietnam runs three trading venues: the Ho Chi Minh City Stock Exchange (HOSE), the Hanoi Stock Exchange (HNX), and UPCOM (an OTC-style board for unlisted public companies). Together they list on the order of 1,600 instruments, with aggregate market capitalisation near 120% of GDP. Liquidity is extremely concentrated: the thirty large caps in the VN30 represent roughly three-quarters of HOSE capitalisation and a large share of turnover, with the index alone trading on the order of VND 12,000–13,000 billion (around USD 500 million) per session in 2025. The VN-Index, base 100 on 28 July 2000, traded in the mid-1,600s in late 2025 after a powerful 2025 rally on the order of 40%+ in price terms, driven substantially by anticipation of the FTSE upgrade.

The practical consequence for a factor researcher is that the investable universe is far smaller than the listed universe. The long tail of UPCOM and small-cap HNX names is thinly traded,

sparsely covered, and prone to stale prices—which biases betas, volatilities, and any signal computed from returns. The reference baskets that matter for implementation are the VN30 (large-cap, institutional-grade), the VNAllShare (a liquidity- and free-float-screened universe covering most of the market’s volume), and the VNDiamond basket (stocks at or near their foreign-ownership ceiling, discussed in the flow family).

2.2 The FTSE upgrade and the marginal investor

FTSE Russell first placed Vietnam on its watchlist in 2018, announced the Frontier-to-Secondary-Emerging reclassification in October 2025, and confirmed it at the March 2026 interim review; the change is effective 21 September 2026 with inclusion phased into 2027. At announced weights, Vietnam is a small slice of the broad benchmarks (roughly 0.2–0.3% of the FTSE Emerging indices), but the passive and active flows tracking those benchmarks are large relative to local float. The upgrade was enabled by concrete reforms—most importantly the removal of the pre-funding requirement for foreign institutional investors (Circular 68/2024), a formal failed-trade handling process, and progress toward a global-broker model.

Two implications for factors. First, the *marginal investor* in the most investable names is shifting toward price-sensitive foreign institutions, which should strengthen valuation-based and quality-based factors in exactly the stocks where foreigners can transact. Second, *index-inclusion demand* becomes a forecastable, calendar-driven flow—a signal in its own right (factor F4).

2.3 Microstructure constraints that bind on implementation

- **Settlement and the KRX system.** The Korea-Exchange-built KRX platform went live on HOSE on 5 May 2025. Cash equities settle T+2 (bonds T+1). The platform enables shorter cycles, market-to-limit (MTL) order handling, and a roadmap to same-day trading; full central counterparty clearing (CCP) is targeted for Q1 2027.
- **No reliable short side (yet).** Covered short selling and single-stock borrow remain restricted/pending. *Market-neutral implementation is therefore constrained:* “long low / short high” factors are, in practice, expressed as long-only underweights or via the (limited) derivatives market. This is the single most important practical caveat in this note.
- **Daily price limits.** Intraday moves are capped at roughly $\pm 7\%$ on HOSE, $\pm 10\%$ on HNX, and $\pm 15\%$ on UPCOM (wider on debut). Limit-locked names produce censored returns, distorting short-horizon reversal and volatility signals.
- **Foreign ownership limits (FOL).** Banks are capped at 30% aggregate foreign ownership (up to 49% for banks involved in restructuring under Decree 69/2025); “conditional” sectors at 50%; many ordinary sectors up to 100%. The KRX system now updates available foreign room in real time at order placement.
- **Pre-funding and access.** Pre-funding for foreign institutions has been removed; omnibus accounts and the global-broker model are being rolled out. Until CCP is fully live, residual settlement and FX-cycle frictions remain.

2.4 Data and survivorship considerations

Three data hazards deserve explicit mention because they silently corrupt factor research in this market: (i) stale prices and zero-return days in the illiquid tail, which deflate measured volatility and bias betas toward zero unless trade-to-trade or Dimson-corrected estimators are used; (ii) accounting-disclosure lags and frequent restatements, which make point-in-time (PIT) fundamental data essential—using as-reported figures with a conservative reporting lag (typically 3 months after fiscal period end) is mandatory to avoid look-ahead; and (iii) corporate-action density—Vietnamese issuers pay stock dividends and conduct rights issues frequently, so all price-based signals must be computed on fully adjusted total-return series.

3 Methodology and conventions

3.1 Investable universe

Unless a factor states otherwise, the working universe is the liquidity- and float-screened set approximated by the VNAllShare: listed equities on HOSE/HNX with (a) a minimum 20-day median traded value above a floor (e.g. VND 1–2 billion/day) to ensure the position is exitable, (b) a minimum free-float and listing history of six months, and (c) exclusion of names under warning, control, or trading suspension. UPCOM is included only for the most liquid names. The universe is re-screened monthly. Capacity-sensitive strategies should additionally cap position weight at a fraction of each name’s average daily traded value (ADTV).

3.2 Standardisation and neutralisation

Each raw factor x_i is transformed into a comparable cross-sectional score before use:

$$z_i = \frac{w(x_i) - \text{median}_j w(x_j)}{\text{MAD}_j w(x_j)}, \quad w(\cdot) = \text{winsorise at the 1st/99th percentile}, \quad (1)$$

using a robust (median / median-absolute-deviation) standardisation to limit the influence of the heavy tails typical of an emerging market. Scores are then **neutralised** by regressing z_i on sector dummies and log(market cap) and retaining the residual:

$$\tilde{z}_i = z_i - (\hat{\alpha} + \hat{\beta}_{\text{size}} \log \text{MktCap}_i + \sum_s \hat{\gamma}_s \mathbb{I}[i \in s]). \quad (2)$$

Sector neutralisation is especially important in Vietnam because the index is dominated by financials and real estate; without it, a “value” or “quality” tilt can collapse into an unintended sector bet.

3.3 Sign convention

Every factor carries an *expected sign*, shown as a chip in its header:

- **LONG HIGH** — the strategy goes long high values of the raw factor (e.g. high earnings yield).
- **LONG LOW** — the strategy goes long low values / would short high values (e.g. low volatility, low asset growth). Given short-sale constraints, this is typically implemented as an underweight.
- **CONTEXT** — the factor is used mainly as a risk, capacity, or conditioning variable rather than a directional alpha; its sign in Vietnam is unstable or context-dependent.

3.4 Evaluation protocol

Factors are judged on a consistent battery: the rank **information coefficient** (Spearman correlation between $\tilde{z}_i$ and forward returns), its mean and t -statistic (the information ratio of the IC series); **quantile spreads** (top-minus-bottom quintile forward return, with monotonicity across quintiles); **turnover and signal decay** (autocorrelation of $\tilde{z}_i$ and the horizon over which the IC persists); and **net-of-cost** performance using a realistic Vietnamese cost stack (brokerage, the ad-valorem transfer tax on sales, bid–ask, and price-impact at the intended size). A factor that looks strong gross but is high-turnover and concentrated in the illiquid tail will not survive this protocol—a common failure mode in this market.

4 The factor catalogue

The thirty factors follow. Each entry gives a precise definition, the economic rationale, and—set apart—a **Vietnam** note flagging local behaviour, plus the **data** required. A star (★) in the text marks factors with particularly strong or particularly fragile evidence in Vietnamese data; read those notes closely. The header chip states the expected long side under the sign convention of Section 3.4.

4.1 Family A --- Value

Value is real in Vietnam—the five-factor model beats the three-factor model and the value factor is non-redundant—but it is concentrated in financials and real estate, so sector neutralisation is essential, and it must be paired with quality to avoid value traps.

V1 Earnings Yield (E/P)

LONG HIGH

$$EP_i = \frac{NI_i^{TTM}}{MktCap_i}, \quad \text{forward variant: } \frac{EPS_i^{fwd}}{P_i}.$$

Rationale. The most basic value signal: stocks cheap relative to trailing (or expected) earnings tend to earn higher subsequent returns. Use the inverse of price-to-earnings to keep the score well-defined when earnings are small.

Vietnam. A core component of the non-redundant value premium. Two cautions: earnings are frequently inflated by one-off gains and revaluations, so pair E/P with the accruals factor (Q4); and because financials dominate the index, neutralise by sector before sorting.

Data. TTM net income (point-in-time, 3-month reporting lag), shares outstanding, price.

V2 Book-to-Price (B/P)

LONG HIGH

$$BP_i = \frac{BookEquity_i}{MktCap_i}.$$

Rationale. The canonical Fama–French HML input. High book-to-market identifies the value cohort.

Vietnam. HML is non-redundant here even after adding profitability and investment, so B/P earns its place. But reported book value is distorted by frequent stock dividends and rights issues, and for banks it reflects regulatory capital—treat the bank sector on its own and neutralise elsewhere.

Data. Latest reported common equity (PIT), price, fully diluted shares.

V3 Sales-to-Price (S/P)

LONG HIGH

$$SP_i = \frac{Revenue_i^{TTM}}{MktCap_i}.$$

Rationale. A value proxy that survives where E/P fails—revenue is positive even for loss-making firms and is harder to manage than earnings.

Vietnam. Particularly useful for the cyclical heavyweights (steel, property, materials) whose earnings swing through the cycle and frequently turn negative, making E/P undefined. A robust complement to V1–V2.

Data. TTM revenue, market capitalisation.

V4 EBITDA-to-EV (operating earnings yield)

LONG HIGH

$$\text{EBITDAEV}_i = \frac{\text{EBITDA}_i^{\text{TTM}}}{\text{EV}_i}, \quad \text{EV}_i = \text{MktCap}_i + \text{NetDebt}_i + \text{Minorities}_i.$$

Rationale. A capital-structure-neutral value measure, comparable across firms with different leverage; the practitioner's preferred value yield.

Vietnam. Valuable given the wide dispersion in leverage between asset-heavy developers and asset-light names. Enterprise value is ill-defined for banks and insurers—exclude financials and score them on P/B (V2) instead.

Data. TTM EBITDA, total debt, cash, minority interest, market cap.

V5 Free Cash Flow Yield

LONG HIGH

$$\text{FCFY}_i = \frac{(\text{CFO} - \text{CapEx})_i^{\text{TTM}}}{\text{EV}_i}.$$

Rationale. Cash-based value: rewards genuine free-cash generation and is resistant to accrual manipulation, the cleanest separation of real value from accounting value.

Vietnam. Powerful but data-constrained—cash-flow statements are less timely and less standardised than the income statement, so apply a coverage filter. Heavy-capex growth and property names screen poorly; sector-neutralise before interpreting.

Data. Operating cash flow, capital expenditure (PIT), enterprise value.

V6 Dividend Yield

LONG HIGH

$$\text{DY}_i = \frac{\text{Cash DPS}_i^{\text{TTM}}}{P_i}.$$

Rationale. An income-plus-quality signal: a high, sustainable cash yield marks mature, cash-generative firms.

Vietnam. Critical adjustment—many Vietnamese firms distribute *stock* dividends, which are not income; count only cash dividends in the numerator and treat stock dividends as splits in the price series. Screen out high yields that arise mechanically from a falling price (value traps) by requiring FCF coverage of the payout.

Data. Trailing cash dividends per share, price; payout and FCF for the sustainability screen.

4.2 Family B --- Quality and profitability

Profitability is one of the best-supported families in Vietnam. The single most important empirical finding for model design is that *operating* profitability dominates other profitability proxies as a factor input.

Q1 Operating Profitability (RMW) ★**LONG HIGH**

$$OP_i = \frac{(\text{Revenue} - \text{COGS} - \text{SG\&A} - \text{Interest})_i^{\text{TTM}}}{\text{BookEquity}_i}.$$

Rationale. The Fama–French RMW (robust-minus-weak) profitability premium: more profitable firms earn higher average returns for a given valuation.

Vietnam. **The strongest profitability proxy in Vietnamese data.** Operating profitability outperforms both cash profitability and ROE as the RMW input and, together with the value factor, makes the largest marginal contribution to the maximum squared Sharpe ratio of the factor set. Make this—not raw ROE—the primary quality input.

Data. Revenue, COGS, SG&A, interest expense, book equity (PIT).

Q2 Gross Profitability**LONG HIGH**

$$GP_i = \frac{(\text{Revenue} - \text{COGS})_i^{\text{TTM}}}{\text{TotalAssets}_i}.$$

Rationale. Novy-Marx’s measure—gross profits scaled by assets—is the profitability figure least polluted by discretionary below-the-line items, and is sometimes called “the other side of value”.

Vietnam. A clean complement to Q1 where SG&A reporting is noisy. Scale by total assets (not equity) so the signal is not contaminated by leverage.

Data. Revenue, COGS, total assets (PIT).

Q3 Return on Equity (ROE)**LONG HIGH**

$$ROE_i = \frac{NI_i^{\text{TTM}}}{\text{BookEquity}_i}.$$

Rationale. The headline profitability ratio, and the one local managers watch most closely.

Vietnam. Works, but is a weaker factor input than operating profitability and is sensitive to leverage and one-off items. Note the structural quirk in banks: the sector earns roughly mid-teens ROE yet trades near book value because the 30% foreign-ownership cap removes valuation-sensitive foreign buyers as marginal price-setters (see F2)—there, high ROE is a *value* signal more than a quality one.

Data. Net income, average book equity (PIT).

Q4 Accruals (earnings quality)**LONG LOW**

$$ACC_i = \frac{(\text{NI} - \text{CFO})_i^{\text{TTM}}}{\text{TotalAssets}_i}.$$

Rationale. Sloan’s accrual anomaly: earnings made up largely of accruals rather than cash are lower quality and predict lower future returns.

Vietnam. Especially relevant because non-cash gains and revaluations are common; low-accrual firms have cleaner, more persistent earnings. Long side is low accruals; the high-accrual short is expressed as an underweight given borrow constraints.

Data. Net income, operating cash flow, total assets (PIT).

Q5 Financial Soundness / Low Leverage

LONG LOW

$$LEV_i = \frac{NetDebt_i}{BookEquity_i}, \quad \text{or distance-to-default } DD_i.$$

Rationale. Lower leverage and greater distance from default reduce distress risk and complement the profitability signal.

Vietnam. Directly addresses a recurring local stress: highly geared property developers with refinancing risk. Underweighting high leverage adds defensiveness through credit cycles. Exclude banks (leverage is their operating model) and score them on capital adequacy instead.

Data. Total debt, cash, equity; for DD also equity volatility and market cap.

Q6 Piotroski F-Score

LONG HIGH

$$F_i = \sum_{k=1}^9 \mathbb{I}[\text{test}_k \text{ passed}] , \quad \text{tests span profitability, leverage/liquidity, efficiency.}$$

Rationale. A nine-point composite of fundamental strength that isolates improving, financially sound firms; historically most effective *within* the cheap cohort.

Vietnam. Use it as a value+quality overlay—apply the F-Score inside the high-B/P quintile to separate healthy value from value traps. Because it aggregates nine binary tests it is robust to single-line accounting noise, a virtue in this disclosure environment.

Data. Two consecutive years of core financial-statement items (PIT).

4.3 Family C --- Investment and growth

The investment leg of the five-factor model carries over to Vietnam; growth as a standalone signal does not behave conventionally and is treated as a style variable.

I1 Asset Growth (CMA)

LONG LOW

$$AG_i = \frac{TotalAssets_{i,t} - TotalAssets_{i,t-1}}{TotalAssets_{i,t-1}}.$$

Rationale. The Fama–French CMA / asset-growth anomaly: conservative (low-investment) firms outperform aggressive ones, as overinvestment tends to destroy value.

Vietnam. Part of the FF5 that out-explains FF3 locally. High asset growth is frequently debt-funded property or capacity expansion that subsequently underperforms; underweight aggressive growers.

Data. Total assets at two consecutive reporting dates (PIT).

G_i = quality-adjusted trailing or expected growth in EPS or revenue.

Rationale. In efficient markets growth carries no positive premium because it is already priced; it is included here as a conditioning style, not a directional alpha.

Vietnam. Sign genuinely regime-dependent. Point-in-time studies have found liquid, growth-oriented large caps outperforming over specific windows—a foreign-flow concentration effect. The safe use is to combine growth with profitability so the model buys *profitable* growth, never growth at any price.

Data. EPS / revenue history, or consensus growth where coverage exists.

4.4 Family D --- Momentum and reversal

This is the family where Vietnam diverges most sharply from developed markets: short-term reversal is strong and dependable, while cross-sectional price momentum is weak and regime-dependent.

M1 Price Momentum (12--1)

CONTEXT

$$\text{MOM}_i = \prod_{\tau=t-12}^{t-2} (1 + r_{i,\tau}) - 1 \quad (\text{skip the most recent month}).$$

Rationale. Jegadeesh–Titman / Carhart momentum: past winners tend to continue over 3–12 month horizons in most markets.

Vietnam. Fragile. Momentum profits appear mainly *after up-markets* and are consistent with behavioural overreaction rather than a stable premium; in many sub-samples they vanish. Do not deploy unconditionally—gate the sleeve on a market-state filter (e.g. index above its long trend) and size it modestly.

Data. Twelve months of adjusted (total-return) prices.

M2 Short-Term Reversal (1-month) ★

LONG LOW

$$\text{REV}_i = -r_{i,t-1 \rightarrow t} \quad (\text{negative of the prior one-month return}).$$

Rationale. Short-horizon reversal driven by liquidity provision and retail overreaction: last month's losers bounce, last month's winners give back.

Vietnam. One of the most robust signals in the market. The one-month reversal is strongly negative, decays by roughly two months and is gone by month three; heavy retail participation amplifies it. Mind the daily price limits, which censor returns at the extremes and can distort the tails of this signal.

Data. Prior one-month return (adjusted).

M3 Earnings Momentum (SUE / PEAD)

LONG HIGH

$$\text{SUE}_i = \frac{\text{EPS}_i^{\text{actual}} - \text{EPS}_i^{\text{expected}}}{\sigma(\text{forecast error})}.$$

Rationale. Post-earnings-announcement drift: prices underreact to earnings surprises, so positive surprises continue to drift up.

Vietnam. Viable but limited by sparse, low-quality analyst coverage. Where consensus is unavailable, proxy “expected” EPS with a seasonal random walk (the year-ago quarter). The drift exists but is noisy—reinforce it with the *direction* of estimate revisions.

Data. Reported quarterly/annual EPS history; analyst consensus where it exists.

M4 52-Week High Proximity

LONG HIGH

$$\text{PH}_i = \frac{P_{i,t}}{\max(P_{i,\tau} : \tau \in [t - 52w, t])}.$$

Rationale. George–Hwang: nearness to the trailing 52-week high predicts continuation, an anchoring effect.

Vietnam. A behavioural signal well-matched to a retail-heavy, anchoring-prone market, and less exposed to the up-market gating that weakens raw momentum—but still confirm regime robustness before sizing it as a standalone.

Data. Fifty-two weeks of adjusted prices.

4.5 Family E --- Risk and volatility

The low-risk anomaly travels reasonably well to emerging markets, and Vietnam's retail dominance makes the "lottery" variants attractive—but every factor in this family is computed from returns, so stale prices and the daily price limits are first-order threats to validity.

R1 Idiosyncratic Volatility

LONG LOW

$$\text{IVOL}_i = \text{std}(\varepsilon_{i,d}), \quad r_{i,d} - r_f = \alpha_i + \beta_i^\top \mathbf{F}_d + \varepsilon_{i,d} \text{ (trailing 1–3 months)}.$$

Rationale. The Ang–Hodrick–Xing–Zhang low-volatility anomaly: stocks with low idiosyncratic volatility earn higher risk-adjusted returns.

Vietnam. Generally effective, but **stale prices in the illiquid tail artificially deflate measured IVOL**, so a naive sort can load on untradeable junk that merely looks calm. Compute on the liquid universe with trade-corrected returns, and cross-check against the zero-return filter (L4).

Data. Daily stock returns and factor returns.

R2 Beta (betting-against-beta)

LONG LOW

$$\beta_i = \frac{\text{Cov}(r_i, r_m)}{\text{Var}(r_m)}, \quad \text{use Dimson (lagged) or Vasicek-shrunk estimates.}$$

Rationale. Frazzini–Pedersen betting-against-beta: low-beta stocks deliver higher risk-adjusted returns as leverage-constrained investors bid up high-beta names.

Vietnam. **Thin trading biases raw betas toward zero**, so an unadjusted low-beta sleeve degenerates into an illiquidity sleeve. Estimate with Dimson lagged-return corrections or Vasicek shrinkage, on liquid names only.

Data. Daily or weekly returns and the market index.

R3 MAX (lottery demand) ★

LONG LOW

$$\text{MAX}_i = \frac{1}{N} \sum_{k=1}^N (k\text{-th highest daily return over the past month}), \quad N \approx 5.$$

Rationale. Bali–Cakici–Whitelaw: stocks with extreme recent positive daily returns attract lottery-seeking buyers, become overpriced, and underperform.

Vietnam. A **strong candidate** in a retail-dominated, lottery-prone market. *But the daily price limits cap single-day returns ($\pm 7\%$ HOSE), censoring MAX.* Adapt the proxy: count the *frequency of limit-up days* over the window rather than raw maxima.

Data. Daily returns over the trailing month (limit-hit flags).

R4 Return Skewness

LONG LOW

$$\text{SKEW}_i = \frac{\frac{1}{D} \sum_d (r_{i,d} - \bar{r}_i)^3}{\left(\frac{1}{D} \sum_d (r_{i,d} - \bar{r}_i)^2 \right)^{3/2}}.$$

Rationale. Investors overpay for positive skew (the lottery payoff), so positively skewed stocks earn lower subsequent returns.

Vietnam. Same retail-preference logic as MAX, and the same censoring caveat. Most useful inside a *lottery composite* that blends MAX (R3), skewness (R4), and turnover (L3) into one “avoid-the-gamble” score.

Data. Daily returns (trailing window).

4.6 Family F --- Liquidity and size

A deliberate reframing for Vietnam: this family is primarily a set of **risk, capacity, and condition-ing controls**, not a dependable alpha source. The textbook size and illiquidity premia are weak or inverted here because foreign capital concentrates in the most liquid names.

L1 Size (log market cap)

CONTEXT

$$\text{SIZE}_i = -\log(\text{MktCap}_i) \quad (\text{small-minus-big orientation}).$$

Rationale. The Fama–French SMB size premium: small caps historically outperform large caps.

Vietnam. Weak. The size effect is not reliably present; foreign and institutional flow favours large caps. Use size mainly as a neutralisation control (Section 3.2), not as a directional bet.

Data. Market capitalisation.

L2 Amihud Illiquidity

CONTEXT

$$\text{ILLIQ}_i = \frac{1}{D} \sum_{d=1}^D \frac{|r_{i,d}|}{\text{VND Volume}_{i,d}}.$$

Rationale. Amihud’s price-impact measure underpins the illiquidity premium: less-liquid stocks should compensate holders with higher expected returns.

Vietnam. Muted or inverted. Because foreign capital crowds into liquid names, illiquid small caps have not reliably paid the textbook premium, and liquid stocks have outperformed over some windows. Treat ILLIQ as a *capacity and risk* control and a conditioning variable—not a long-illiquid alpha.

Data. Daily returns and daily VND traded value.

L3 Share Turnover

LONG LOW

$$\text{TURN}_i = \frac{\overline{\text{daily volume}_i}}{\text{shares outstanding}_i} \quad (\text{trailing}).$$

Rationale. High turnover signals attention, speculation, and frequently overvaluation; it predicts lower future returns and also proxies liquidity.

Vietnam. Effective for flagging retail-driven speculative names; underweighting high-turnover stocks is defensive and dovetails with the lottery composite (R3–R4).

Data. Trading volume and shares outstanding.

$$\text{LOT}_i = \frac{\#\{\text{days with } r_{i,d} = 0\}}{D} \quad (\text{Lesmond-Ogden-Trzcinka}).$$

Rationale. A liquidity and transaction-cost proxy computable without intraday data.

Vietnam. Indispensable as a **tradability filter and stale-price detector**: a high zero-return proportion marks thin trading and unreliable risk estimates. Use it to clean the universe and to down-weight names where return-based signals (R1–R4) cannot be trusted.

Data. Daily returns (zero-day count); quoted spreads where available.

4.7 Family G --- Vietnam-specific flow and microstructure

This family has no clean developed-market analogue, and it is where a local researcher's edge concentrates. All four exploit structural features of the Vietnamese market: per-stock foreign-flow disclosure, statutory ownership caps, pervasive state ownership, and the impending index upgrade.

F1 Foreign Net Flow ★

LONG HIGH

$$\text{FNF}_i = \frac{(\text{net foreign buy value})_i^{[t-h, t]}}{\text{MktCap}_i} \quad (\text{level or short-horizon momentum}).$$

Rationale. Foreign institutions are persistent, comparatively informed buyers; their per-stock net flow carries information and exerts price pressure, predicting near-term cross-sectional returns.

Vietnam. A genuinely local alpha—foreign buy/sell value is published per stock and watched closely. Important nuance: 2025 saw heavy aggregate net foreign *selling* (on the order of USD 4.7 billion on HOSE) even during the rally, so use FNF as a *cross-sectional* signal (which names foreigners favour) rather than a market-timing tool. Expect its relevance to rise through the FTSE-inclusion window.

Data. Daily per-stock foreign buy/sell value (exchange-published).

F2 Foreign Room Remaining

CONTEXT

$$\text{ROOM}_i = \text{FOL}_i - \text{Foreign Ownership}_i \quad (\text{as \% of shares; "room-full" indicator}).$$

Rationale. When foreign room is exhausted, foreigners cannot be the marginal buyer: such names trade at a foreign premium via negotiated deals and lose the upgrade-flow tailwind; conversely, a cap increase can unleash pent-up demand.

Vietnam. A **defining microstructure factor**. Room-full large caps (several have sat at 100% foreign ownership for years) behave differently from open-room peers, and the 30% bank cap structurally depresses bank valuations by leaving local retail as the marginal price-setter. Use ROOM two ways: (i) as a *gate* on whether the foreign-flow and inclusion signals (F1, F4) can physically act on a name, and (ii) as an *event* signal when caps are raised. The VNDiamond basket institutionalises the room-full cohort.

Data. Current foreign ownership %, statutory FOL by sector/firm.

$$SOE_i = \text{government} / \text{SCIC ownership stake (\%)}$$

Rationale. Ownership structure shapes governance, free float, and the probability of value-unlocking corporate events (divestment, equitisation).

Vietnam. State ownership is pervasive. High-SOE names carry governance discounts and thin free float, yet are also divestment candidates whose re-rating can be large when the state sells down. Use as a conditioning/event variable; evidence that factor premia differ across state-ownership cohorts further motivates treating it explicitly.

Data. Ownership registry; the official divestment / equitisation calendar.

$$INCL_i = \frac{\hat{w}_i \times \widehat{AUM}_{\text{tracking}} \wedge ROOM_i}{ADTV_i} \quad (\text{expected passive demand in days of volume}).$$

Rationale. Index reconstitutions generate forecastable, price-insensitive passive demand; positioning ahead of the mechanical flow is a classic event alpha.

Vietnam. Highly topical. FTSE Secondary-Emerging inclusion—effective 21 September 2026 and phased into 2027—creates a calendar of mechanical buying concentrated in the largest, room-available names. Rank stocks by estimated inclusion weight times tracking AUM, capped at available foreign room (F2), scaled by ADTV. The signal decays after the effective dates; combine with foreign net flow (F1).

Data. Projected index weights, tracking-AUM estimates, foreign room, ADTV.

5 Vietnam evidence: a comparative synthesis

The table below condenses the catalogue into a single view: the long side under our sign convention, a qualitative read of the strength of the evidence *in Vietnamese data*, and the single most important caveat for each factor. “Strong” marks factors with robust local support or a clear structural basis; “inverted/weak” marks factors where the developed-market prior does not hold here. Use it as a triage map when deciding which sleeves to build first.

ID	Factor	Long side	VN evidence	Primary caveat
A. Value				
V1	Earnings yield (E/P)	High	Moderate–strong	One-off earnings; financials tilt
V2	Book-to-price (B/P)	High	Moderate–strong	Book distortions; banks separate
V3	Sales-to-price (S/P)	High	Moderate	Best for cyclicals / negative earnings
V4	EBITDA/EV	High	Moderate	Exclude financials (EV undefined)
V5	FCF yield	High	Moderate	CF data lag; capex-heavy names
V6	Dividend yield	High	Moderate	Cash-only DPS; value traps
B. Quality and profitability				
Q1	Operating profitability *	High	Strong	Best RMW proxy—prioritise
Q2	Gross profitability	High	Moderate	Scale by assets, not equity
Q3	Return on equity	High	Moderate	In banks, a value (not quality) signal
Q4	Accruals	Low	Moderate	Non-cash gains common
Q5	Low leverage / soundness	Low	Moderate	Property distress; ex-banks
Q6	Piotroski F-score	High	Moderate–strong	Apply within the cheap cohort
C. Investment and growth				
I1	Asset growth (CMA)	Low	Moderate	Debt-funded expansion
I2	Earnings/sales growth	Context	Regime-dependent	Combine with profitability
D. Momentum and reversal				
M1	Price momentum (12–1)	Context	Weak / regime	Gate on market state
M2	Short-term reversal *	Low prior	Strong	Price-limit censoring at tails
M3	Earnings momentum (SUE)	High	Moderate	Sparse coverage; proxy expected EPS
M4	52-week high	High	Moderate	Confirm regime robustness
E. Risk and volatility				
R1	Idiosyncratic volatility	Low	Moderate–strong	Stale-price deflation bias
R2	Beta (low-beta)	Low	Moderate	Use Dimson / shrinkage beta
R3	MAX (lottery) *	Low	Strong	Limits censor max; use limit-up freq
R4	Return skewness	Low	Moderate	Part of a lottery composite
F. Liquidity and size				
L1	Size	Context	Weak	Use for neutralisation
L2	Amihud illiquidity	Context	Weak / inverted	Capacity control, not alpha
L3	Turnover	Low	Moderate	Speculative-name flag
L4	Zero-return / spread	Context	Filter	Stale-price detector
G. Vietnam-specific flow and microstructure				
F1	Foreign net flow *	High	Strong (local)	Cross-sectional, not market timing
F2	Foreign room remaining	Context	Structural	Gates F1/F4; banks capped 30%
F3	State / SOE ownership	Context	Event / governance	Divestment as catalyst
F4	Index-inclusion demand *	High (event)	Topical / strong	Decays after effective date

6 Composite construction, capacity, and costs

6.1 Combining factors

Two defensible routes. The **composite-score** route forms category scores by averaging the neutralised $\tilde{z}$ of the factors in each family (so no single family dominates by count), then combines categories with weights set by conviction or by a risk-parity scheme on the category return series; this is transparent and robust. The **multi-factor cross-sectional regression** route estimates factor returns by regressing forward returns on all $\tilde{z}_i$ jointly (a Fama–MacBeth or characteristic-portfolio approach), which handles factor correlation explicitly but is more sensitive to estimation noise in a market with this many illiquid names. For an initial Vietnam model the composite route is recommended, with the regression used as a diagnostic on factor redundancy.

A practical prior on weights, given the evidence table: overweight the *quality* (especially Q1), *value*, and *Vietnam-flow* (F1, F4) families; treat *momentum* cautiously and conditionally (favour M2 over M1); and demote *liquidity/size* to controls. Diversify across families that are weakly correlated—value and reversal, for instance, are largely orthogonal here.

6.2 The short-sale constraint and what it means for design

Because covered short selling and stock borrow remain restricted, the **LONG LOW** factors cannot be expressed as clean shorts. In practice the model is run as a **long-only tilt**: rank by the composite and overweight the top cohort versus a benchmark (VN30 or VNAllShare) while underweighting or excluding the bottom. This halves the theoretical breadth of a long–short design and means defensive factors (low volatility, low leverage, low accruals) act mainly through what the portfolio *avoids*. Where a hedge is required, the (limited) index-futures market is the only liquid instrument.

6.3 Capacity and rebalancing

Capacity is set by ADTV, not by market cap: cap each position at a small fraction (e.g. 5–10%) of its 20-day median traded value so the book can be built and unwound within a few sessions. Because the dominant alpha (reversal) decays within roughly a month while fundamental factors decay over quarters, a sensible cadence is **monthly** rebalancing of the fast sleeves and **quarterly** refresh of fundamentals, with turnover penalised in the optimiser. Concentrate live capital in the VNAllShare-screened universe; the illiquid tail is where gross backtests look best and net results disappoint.

6.4 The Vietnamese cost stack

Model costs explicitly: broker commission (broadly 0.1–0.15% per side), the ad-valorem *securities transfer tax levied on the sell side* (a fixed percentage of sale proceeds regardless of profit), bid–ask spread, and price impact at the intended size. The sell-side transaction tax in particular penalises high-turnover signals more than in markets with symmetric, profit-based costs, reinforcing the case for turnover control and for favouring the slower fundamental sleeves over hyperactive trading of the reversal signal.

7 Implementation pitfalls specific to Vietnam

1. **Look-ahead through disclosure timing.** Use point-in-time fundamentals with a conservative reporting lag; back-filled or restated figures inflate every fundamental factor.
2. **Stale prices and zero-return days** deflate volatility and bias betas toward zero, contaminating the entire risk family—filter with L4 and use trade-corrected estimators.

3. **Price-limit censoring** truncates short-horizon returns, distorting reversal (M2), MAX (R3), and skewness (R4); flag limit-locked observations and adapt the lottery proxies to limit-hit frequencies.
4. **Sector concentration.** Financials and real estate dominate; without sector neutralisation a value or quality tilt becomes an unintended macro bet on banks and developers.
5. **Survivorship and corporate actions.** Include delisted names and adjust rigorously for the frequent stock dividends and rights issues, or price-based signals are silently wrong.
6. **Foreign room at execution.** A buy signal on a room-full name (F2) may be unfillable for a foreign book; screen tradability *before* sizing, not after.
7. **Thin analyst coverage** undercuts consensus-based signals (M3); proxy expectations with seasonal models where coverage is absent.
8. **Regime dependence of momentum** (M1): an unconditional long-momentum sleeve is unsafe; gate it on market state.
9. **Crowding** into the VN30 / VNDiamond cohort: the most investable names are also the most owned by foreigners and ETFs, so apparent factor exposure there can be a crowded-positioning exposure—monitor and diversify.

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Appendix A --- Notation and abbreviations

ADTV	Average daily traded value (in VND); the binding capacity constraint.
CCP	Central counterparty clearing; full operation targeted Q1 2027.
CFO	Cash flow from operations.
EV	Enterprise value = market cap + net debt + minorities – cash.
FOL	Foreign ownership limit (statutory cap on aggregate foreign holding).
HML / SMB / RMW / CMA	Fama–French value / size / profitability / investment factors.
HOSE / HNX / UPCOM	The three Vietnamese trading venues.
IC / IR	Information coefficient (rank correlation of signal with forward return) / information ratio.
MAD	Median absolute deviation (robust scale used in standardisation).
PEAD / SUE	Post-earnings-announcement drift / standardised unexpected earnings.
PIT	Point-in-time data (as known on the date, with reporting lag).
TTM	Trailing twelve months.
VN30 / VNAIIShare / VNDiamond	Large-cap / broad liquid-universe / room-full reference baskets.

End of document. This is a research specification prepared for internal quantitative use; it is not investment advice, and the qualitative “evidence” assessments reflect the cited literature and market structure as of mid-2026, not a guarantee of future factor performance.